

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Discuss the technique of emulsion polymerization in detail.
- Q.24 Explain the factors which influences glass transition temperature on properties of polymer materials.
- Q.25 Explain the reaction mechanism of ionic polymerization with example.

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1st Sem / Plastic Technology

**Subject : Introduction to Polymer Science and Technology
/ Polymer Science and Technology/ Poly. Sc. & Tech. I**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Incompressible fluids are
- a) Density does not change with time
 - b) Density changes with time
 - c) Can not say
 - d) None of these
- Q.2 Amorphous polymers are
- a) Regular in shape b) Irregular in shape
 - c) Can not say d) None of these
- Q.3 Newtonian fluids are those which follow
- a) Newton's law of viscosity
 - b) Hooks law
 - c) Pascal Law
 - d) None of these

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Q.4 SI unit of kinematics is

- a) m^2/s b) kg/m^3
- c) m/s d) m/s^2

Q.5 For turbulent flow, the value of Reynolds number is

- a) $N_{\text{Re}} > 4000$ b) $N_{\text{Re}} < 2000$
- c) $N_{\text{Re}} = 3000$ d) All of these

Q.6 The sequence of polymerization reaction is

- a) Initiation, Propagation & Termination
- b) Initiation, Termination & Propagation
- c) Termination, Propagation & Initiation
- d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define crystalline polymer.

Q.8 Define Newtonian fluids.

Q.9 Define Viscosity.

Q.10 Define ideal fluid with example.

Q.11 Define free radical.

Q.12 Define visco-elasticity.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Discuss the Maxwell model of visco-elastic materials.

Q.14 Write a short note on amorphous behaviour of polymers.

Q.15 Explain one experimental technique for measurement of glass transition temperature (T_g)

Q.16 How is copolymers importance to us?

Q.17 Define copolymers and their types.

Q.18 Write short note on polymer blends.

Q.19 Explain glass transition temperature (T_g) and its importance.

Q.20 Explain the process of bulk polymerization.

Q.21 Explain the concept of gel permeation chromatography for measurement of molecular weight of polymers.

Q.22 Describe the reaction mechanism of addition polymerization.

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